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Comparative Analysis of Bias Mitigation Techniques in Supervised Machine Learning

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Comparative Analysis of Bias Mitigation Techniques in Supervised Machine Learning"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

Automated decision-making systems are now a critical part of high-stakes domains such as hiring, credit allocation, clinical triage, and criminal risk assessment. However, the outcomes they produce often reinforce the unfairness embedded in their training data. This report presents a methodical and direct comparison of six bias mitigation strategies sourced from three separate intervention stages in the supervised learning pipeline: pre-processing, in-processing, and post-processing.

Three classifiers of varying structural complexity — Logistic Regression, Decision Tree, and Random Forest — were employed to evaluate two well-known benchmark datasets: the Adult Income dataset and the COMPAS recidivism dataset. Fairness is assessed using standard accuracy and macro-F1 measures, alongside Demographic Parity Difference (DPD) and Equal Opportunity Difference (EOD).

The findings demonstrate that all interventions considerably reduce group-level disparity, although the magnitude of the accuracy-fairness trade-off varies substantially between intervention stages. In-processing methods achieve the greatest bias reductions at a higher prediction accuracy cost, while pre-processing techniques yield small but consistent fairness improvements with minimal performance decrease. Post-processing techniques serve as a practical middle ground, enabling threshold adjustments without requiring model retraining. These findings provide organized evidence for selecting mitigation solutions that satisfy specific accuracy and fairness requirements.

Keywords: Fairness in machine learning, bias mitigation, supervised learning, ethical AI, fairness metrics, demographic parity, equal opportunity, algorithmic accountability.

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INTRODUCTION

1.1 Background and Motivation

Machine learning systems are being increasingly deployed in critical decision-making domains. Hiring platforms score résumés, lending algorithms assess creditworthiness, clinical tools evaluate patient risk, and recidivism prediction tools help determine bail outcomes — all at a speed and scale that no human reviewer can match. However, speed does not guarantee fairness. When a model learns from decades of historical records that reflect unequal treatment, it does not merely describe the past; it encodes it, projects it into the future, and operationalises it at industrial throughput.

Outcomes that once depended on deliberate human bias can now arise automatically through weights and thresholds that are entirely opaque to the individuals most affected by them. This shift from explicit to implicit discrimination makes the problem more difficult to detect, challenge, and correct.

1.2 Problem Statement

Bias can enter machine learning systems at multiple stages and through multiple pathways, and these pathways often compound one another. Training corpora that underrepresent certain demographic groups produce models that are poorly calibrated for those populations. Proxy variables — features that are statistically correlated with protected attributes without naming them directly — allow models to recreate sensitive distinctions even when those attributes are intentionally excluded. Label noise, arising from biased historical human judgements, propagates errors into future predictions. The result is a class of systems that can achieve high overall accuracy metrics while generating consistently adverse outcomes for specific subgroups.

1.3 Objectives

This report addresses the following research objectives:

1. To provide a single experimental benchmark comparing six representative bias mitigation techniques — two for each pipeline stage — assessed under identical conditions.
2. To analyse the accuracy-fairness trade-off across three classifier families (Logistic Regression, Decision Tree, and Random Forest) and two benchmark datasets.
3. To provide empirical evidence supporting informed selection of bias mitigation techniques based on deployment constraints, accuracy budgets, and fairness requirements.
4. To contribute a practitioner-focused decision framework that can guide technique selection in real-world deployments.

1.4Scope of the Report

This project is scoped to supervised binary classification tasks. The study covers six bias mitigation techniques spanning pre-processing, in-processing, and post-processing stages of the machine learning pipeline. Two publicly available benchmark datasets — the Adult Income dataset and the COMPAS dataset — are used. The evaluation is restricted to group fairness metrics (Demographic Parity Difference and Equal Opportunity Difference) alongside standard predictive performance metrics (accuracy and macro-F1 score).

LITERATURE REVIEW

2.1 Conceptual Foundations of Fairness

A fundamental tension emerged in the formal study of fairness: the intuitive desire to treat individuals equally does not always translate into a single, mathematically consistent concept. Barocas, Hardt, and Narayanan [1] distinguished between group fairness, which requires statistical parity across demographic categories, and individual fairness, which requires that similar individuals receive similar treatment. This conceptual framework served as the foundation for much subsequent work. The metrics that emerged — Equal Opportunity Difference and Demographic Parity Difference — are now standard tools for assessing classifier equity, and this study employs both.

2.2 Pre-Processing Approaches

Calders, Kamiran, and Pechenizkiy [2] demonstrated that instance-level weights calibrated to the joint distribution of sensitive attributes and outcome labels could significantly reduce discriminatory patterns in Naïve Bayes classifiers without altering the underlying data structure. This finding established reweighing as one of the most practical tools in the fairness toolkit. Kamiran and Calders [3] expanded this inquiry to resampling techniques, showing that selectively oversampling and undersampling based on group-outcome combinations enhanced fairness on real income-prediction benchmarks, establishing data-level adjustments as a reliable first line of defence against inherited bias.

2.3 In-Processing Approaches

More sophisticated mechanisms were introduced by the in-processing literature. By adding covariance-based fairness constraints to the logistic regression objective, Zafar et al. [4] turned what had previously been a post-hoc concern into a first-class optimisation target. Their method demonstrated that accuracy and fairness need not be viewed as mutually exclusive goals — in certain problem geometries, model-level constraints only marginally reduce performance. Agarwal et al. [5] cast fair classification as a constrained optimisation problem solvable through a sequence of standard weighted reductions, making in-processing techniques

accessible to practitioners already familiar with off-the-shelf learning libraries. Adversarial debiasing involves training an auxiliary classifier that attempts to predict sensitive attributes from the primary model's intermediate representations, penalising the main model for encoding group-identifiable patterns [4].

2.4 Post-Processing Approaches

Post-processing corrections, methodically introduced by Hardt, Price, and Srebro [6], operate solely on predicted scores rather than the model itself — making them suitable when retraining is costly or the underlying model is a proprietary black box. By identifying group-specific decision thresholds that minimise error-rate parity violations, their equalized odds post-processor achieves fairness without affecting training data or model parameters. Pleiss et al. [7] enhanced this line of work by adding score calibration to the threshold adjustment, ensuring that predicted probabilities maintain consistent probabilistic meaning across demographic groups.

2.5 Gaps in Existing Research

Despite this wealth of individual contributions, research has rarely directly compared the three intervention families in controlled experimental settings. Bellamy et al. [8] provided a comprehensive assessment through the AI Fairness 360 toolbox; however, their evaluation covered a narrower range of classifiers and lacked the cross-dataset consistency that would enable confident generalisations. The current study addresses that gap by applying a consistent experimental framework to six techniques, two datasets, and three classifiers.

METHODOLOGY

3.1 Dataset Selection

Two benchmark datasets were selected because they represent real-world classification contexts where bias has observable societal consequences, are well-understood in the fairness research community, and are publicly accessible for reproducibility.

The Adult Income Dataset comprises 48,842 records from the 1994 U.S. Census, obtained from the UCI Machine Learning Repository. Each record includes 14 economic and demographic features; the binary outcome indicates whether annual income exceeds $50,000. Gender was selected as the sensitive attribute, reflecting verified pay inequalities in the underlying population.

The COMPAS Dataset contains 7,214 defendant records from Broward County, Florida, made public by ProPublica following its 2016 investigation. The target variable is a binary indicator of re-arrest within two years of release. Race was identified as the sensitive attribute, consistent with the extensively documented racial disparities in COMPAS risk scores. Incorporating both datasets ensures that findings are not limited to a particular domain, type of sensitive attribute, or degree of baseline disparity.

3.2 Baseline Models

Three classifiers were selected to establish a baseline: Logistic Regression, Decision Tree, and Random Forest. This selection covers a range of model complexities — from linear boundary estimation to ensemble tree averaging — enabling the study to determine whether bias mitigation techniques generalise across structurally different architectures or perform selectively. All classifiers were implemented using Scikit-learn with hyperparameters tuned via five-fold cross-validation on the training partition only.

3.3 Bias Mitigation Techniques

Six intervention strategies were examined, two from each pipeline stage:

Pre-Processing:

- Reweighing: Assigns scalar weights to training instances based on their sensitive attribute and outcome label combination, adjusting the effective distribution without altering actual feature values.
- Disparate Impact Remover: Modifies non-sensitive feature distributions within each group to reduce the statistical correlation between feature values and group membership.

In-Processing:

- Adversarial Debiasing: Trains an adversarial network alongside the primary classifier; the primary model is penalised whenever the adversary successfully infers the sensitive attribute from its representations.
- Prejudice Remover: Adds a regularisation term to the logistic loss function that penalises mutual information between predictions and sensitive attributes.

Post-Processing:

- Equalized Odds Post-Processing: Determines group-specific decision thresholds by solving a linear program that minimises violations of the equalized odds criterion.
- Calibrated Equalized Odds: Extends equalized odds post-processing by additionally enforcing score calibration, ensuring predicted probabilities carry consistent empirical meaning across groups.

3.4 Evaluation Metrics

Model performance was evaluated using accuracy and macro-averaged F1-score. Macro-F1 weights each class equally rather than aggregating across instances, making it more informative under class imbalance — a condition present in both datasets. Fairness was quantified using:

- Demographic Parity Difference (DPD): The absolute difference in positive prediction rates between advantaged and disadvantaged groups. A DPD of zero indicates both groups receive favourable outcomes at equal rates.
- Equal Opportunity Difference (EOD): Restricts attention to qualified individuals and measures the difference in true positive rates across groups, assessing whether the system is equally effective at identifying deserving candidates in each group.

3.5Experimental Setup

Each experiment was implemented in Python 3.10. Baseline classifiers were sourced from Scikit-learn; bias mitigation methods were sourced from IBM's AI Fairness 360 toolkit [8]. Stratified sampling was applied to divide the data into training and test sets in an 80:20 ratio, maintaining class proportions across splits. Hyperparameters were selected using five-fold cross-validation applied exclusively to the training partition. Fixed random seeds were used across all trials to ensure reproducibility.

TOOLS AND TECHNOLOGIES

4.1 Programming Language

Python 3.10 was the primary programming language used for all experiments. Python's extensive ecosystem for machine learning, data processing, and fairness research made it the natural choice. Its interpretability and widespread adoption in the research community also ensure that the experiments can be replicated and extended by others.

4.2 Machine Learning Library – Scikit-learn

Scikit-learn was used for implementing the three baseline classifiers: Logistic Regression, Decision Tree, and Random Forest. Scikit-learn provides consistent, well-documented APIs for model training, cross-validation, and evaluation, enabling standardised comparison across classifier architectures. Its stratified K-fold utility was used for hyperparameter selection.

4.3 Fairness Toolkit – IBM AI Fairness 360 (AIF360)

The IBM AI Fairness 360 (AIF360) toolkit was the central tool for implementing all six bias mitigation techniques. AIF360 provides a unified interface for pre-processing, in-processing, and post-processing fairness interventions, along with built-in metrics for measuring bias including Demographic Parity Difference and Equal Opportunity Difference. Its design philosophy of separating mitigation logic from model training logic made it ideal for controlled comparative experiments.

4.4 Data Sources

The Adult Income Dataset was obtained from the UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/adult). The COMPAS Recidivism Dataset was obtained from the ProPublica GitHub repository, which made the dataset public following their 2016 investigative analysis of racial bias in algorithmic criminal risk assessment.

4.5 Development Environment

Experiments were conducted in a Jupyter Notebook environment using Anaconda distribution. Version control was managed using Git. All library versions were pinned in a requirements file to ensure full reproducibility of results.

RESULTS AND DISCUSSION

5.1 Baseline Performance – Adult Income Dataset

Table I presents accuracy, F1-score, DPD, and EOD across baseline classifiers on the Adult Income dataset, without any fairness intervention applied. The Random Forest achieves the highest predictive performance (86.2% accuracy, F1 = 0.831), but also produces the largest fairness disparity (DPD = 0.197). This reveals a critical paradox: the model with the best predictive quality is simultaneously the most unfair. High accuracy does not guarantee equitable outcomes across demographic groups.

Table I: Baseline Model Performance – Adult Income Dataset

Model	Accuracy (%)	F1-Score	DPD	EOD
Logistic Regression	82.4	0.798	0.149	0.132
Decision Tree	83.7	0.811	0.163	0.147
Random Forest	86.2	0.831	0.197	0.178

5.2 Impact of Bias Mitigation – Logistic Regression

Table II summarises the results for each of the six mitigation techniques applied to Logistic Regression on the Adult Income dataset. All six methods reduce both DPD and EOD relative to the untreated baseline, confirming that the interventions function as intended. However, the degree of bias reduction and the associated performance cost differ substantially across the three intervention stages.

Table II: Post-Mitigation Results – Logistic Regression (Adult Income)

Technique	Accuracy (%)	F1-Score	DPD	EOD
Reweighing	81.9	0.787	0.092	0.083
DI Remover	80.3	0.774	0.078	0.071
Adversarial Debiasing	79.1	0.762	0.043	0.039

Prejudice Remover	78.8	0.759	0.051	0.046
Eq. Odds Post-Processing	80.7	0.776	0.067	0.062
Calibrated Eq. Odds	81.2	0.781	0.071	0.065

5.3 Fairness–Accuracy Trade-off Analysis

Pre-processing techniques (Reweighing and DI Remover) reduce DPD to 0.092 and 0.078, respectively, at a cost of fewer than two percentage points of accuracy. These strategies modify the weighting or marginal distribution of training instances without altering the classifier's learning dynamics, which is consistent with the small accuracy penalty observed.

In-processing techniques achieve the most dramatic bias reductions — DPD drops to 0.043 (Adversarial Debiasing) and 0.051 (Prejudice Remover) — but at the cost of 3.1 and 3.4 accuracy points respectively. This trade-off reflects the fundamental tension of penalising group-correlated learning signals while simultaneously optimising for a discriminative objective.

Post-processing techniques occupy the middle ground. Calibrated Equalized Odds (DPD = 0.071, accuracy = 81.2%) maintains slightly more predictive value than its uncalibrated counterpart (DPD = 0.067, accuracy = 80.7%), suggesting that the calibration constraint provides effective regularisation.

5.4 COMPAS Dataset Observations

A consistent trend across the COMPAS dataset is that DPD values remain persistently high (above 0.25) across all classifiers and all mitigation techniques. This finding indicates that the racial disparities in the COMPAS scoring context are deeply embedded in the data structure, reflecting pre-algorithmic societal inequities that technical interventions alone cannot fully resolve. Both datasets show similar response patterns to the same classes of adjustments, indicating that these disparities are not random artefacts but are predictable and structured.

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project addressed a practical question central to responsible AI deployment: which family of bias mitigation approaches should a practitioner prioritise given a specified deployment environment and an acceptable accuracy budget? The findings, derived from six techniques assessed across two datasets and three classifiers under consistent experimental conditions, provide a formal basis for that decision.

Pre-processing methods emerge as the most accessible starting point. Their small accuracy penalties, compatibility with standard training pipelines, and independence from underlying model architecture make them well-suited for production settings where retraining is costly and model modifications carry compliance overhead. Reweighing in particular provides a reliable first step for organisations beginning their fairness journey.

In-processing techniques perform best in environments where fairness is a core design requirement and where regulatory frameworks permit or require slight accuracy reductions. Their strong bias reductions come at quantifiable and predictable costs. Post-processing methods are uniquely valuable for retrofitting fairness into deployed systems without interfering with live pipelines, since they require no retraining.

These findings reinforce that algorithmic fairness is not a single problem with a single solution. Technique selection must be guided by deployment constraints, accuracy tolerances, interpretability requirements, computational budgets, and the nature of the sensitive feature. As legal frameworks such as the EU AI Act increasingly mandate bias evaluation for high-risk AI applications, the type of comparative empirical evidence generated by this study becomes practically essential.

6.2 Future Scope

Several promising directions for future research emerge from this work:

- Intersectional Fairness: Extending the analysis to intersectional sensitive attributes (e.g., race and gender simultaneously) to evaluate how mitigation effectiveness changes under multi-dimensional protected group analysis.
- Deep Learning and LLM Pipelines: Evaluating the efficacy of these mitigation strategies when applied to deep neural network architectures and fine-tuning pipelines for large language models, where bias manifestations are significantly more complex.
- Acceptable Trade-off Standards: Developing domain-specific ethical standards for acceptable accuracy-fairness trade-offs that can be integrated into model approval workflows, particularly in high-stakes regulated environments.
- Temporal Bias Drift: Investigating how bias mitigation effectiveness degrades over time as data distributions shift, and developing adaptive mechanisms for maintaining fairness in continuously updated models.

Ultimately, bias in machine learning reflects bias in the world that produced the data. Algorithmic interventions, however rigorous, are necessary but not sufficient. The most resilient path to equitable automated decision-making requires both the technical rigour demonstrated here and a sustained commitment to the institutional and social conditions that govern what training data is collected, how labels are assigned, and whose experiences are fairly represented.

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